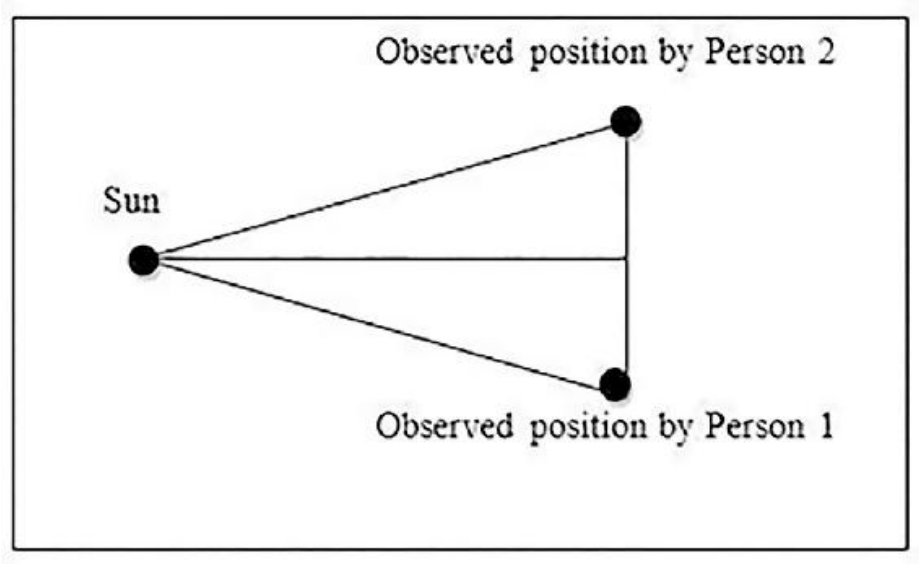
So, if a simplistic star chart were to be drawn of the surrounding fixed points in the sky—like the moon or Venus, for example—each person observing the sun must mark the observed position of the sun. If the observers are 10 meters apart, there will be a ten-meter disconnect in their recorded object locations relative to the other markers in the star chart. Therefore, a new triangle is formed as if the first has been flipped over, leaving the sun at the same spot.



The distance between the observers and their respective observed positions of the sun is the same. Since the calculation must be performed on a right triangle, the model is split in half, making the shortest line of the triangle only half the distance that was initially between the two observers. Using a secondary calculation, astronomers can find the precise angular measurement for that most acute angle, then use the sine ratio to determine the length of the other two lines. Therefore, the sun and the earth are separated by nearly 150 million kilometers, or 93 million miles.

Aryabhata was by no means the only mathematical innovator to come from India. Another important contributor to the field was Brahmagupta, an Indian astronomer and mathematician of the 6th century. His work is evidenced deep inside the sandstone city of Fort Gwalior in central India where archaeologists found the first written proof of what was perhaps India's greatest addition to mathematics: the zero. Probably in use since the 5th century, the concept of zero was a difficult one to grasp for even the brightest minds of the age—and yet it was a digit without which the